

TYPES OF NETWORK

Computer networks are classified by their geographic scope, with the main types being:

- Personal Area Network (PAN)
- Metropolitan Area Network (MAN)
- Wide Area Network (WAN)
- Wireless LANs (WLANs)
- Campus Area Networks (CANs)
- Virtual Private Networks (VPNs)
- Enterprise Private Network (EPN)

Primary Network Types

1. Personal Area Network (PAN).
2. Local Area Network (LAN):
3. Metropolitan Area Network (MAN):
4. Wide Area Network (WAN):

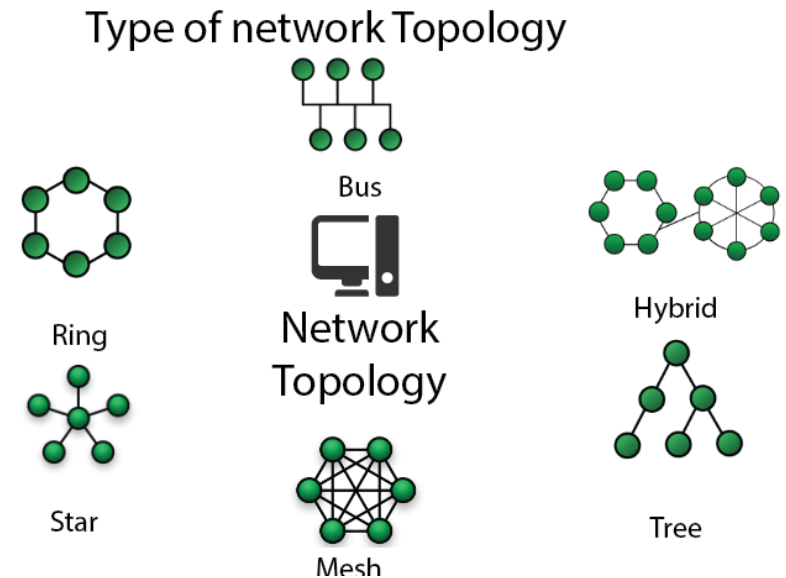
NETWORK TOPOLOGIES

Network topology refers to the physical or logical arrangement of nodes and connections in a network.

Why these topologies?

Essential for designing efficient, scalable, and reliable network infrastructures.

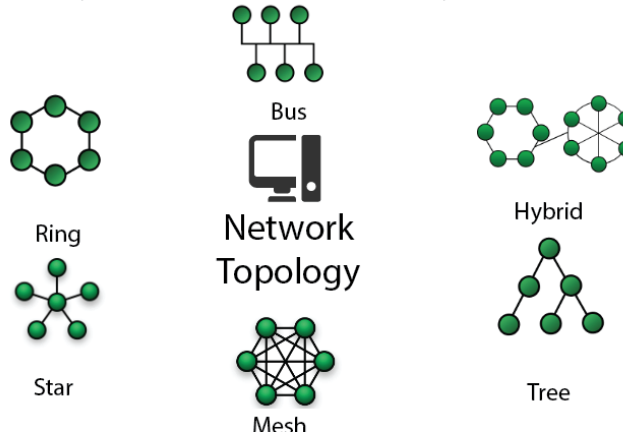
Wired or Wireless Network follows topology?



NETWORK TOPOLOGY

Definition, Types, Advantages & Applications

Type of network Topology



WHAT IS NETWORK TOPOLOGY?

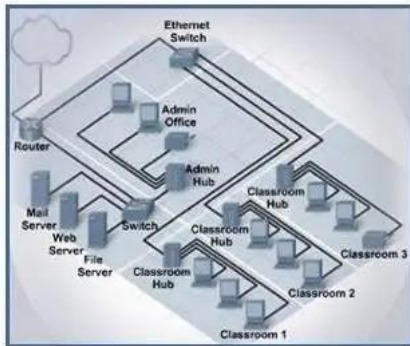
Network topology refers to the arrangement of nodes, links, and devices in a network.

It defines how devices are connected and how data flows between them.

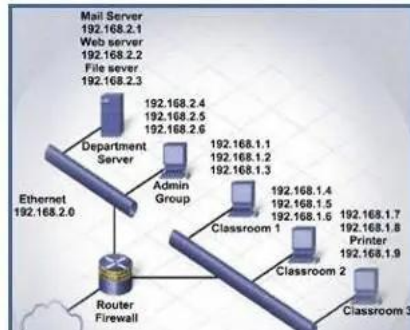
CATEGORIES OF NETWORK TOPOLOGY

- Physical Topology: Actual physical layout of devices
- Logical Topology: Path data follows irrespective of physical layout

CATEGORIES OF NETWORK TOPOLOGY



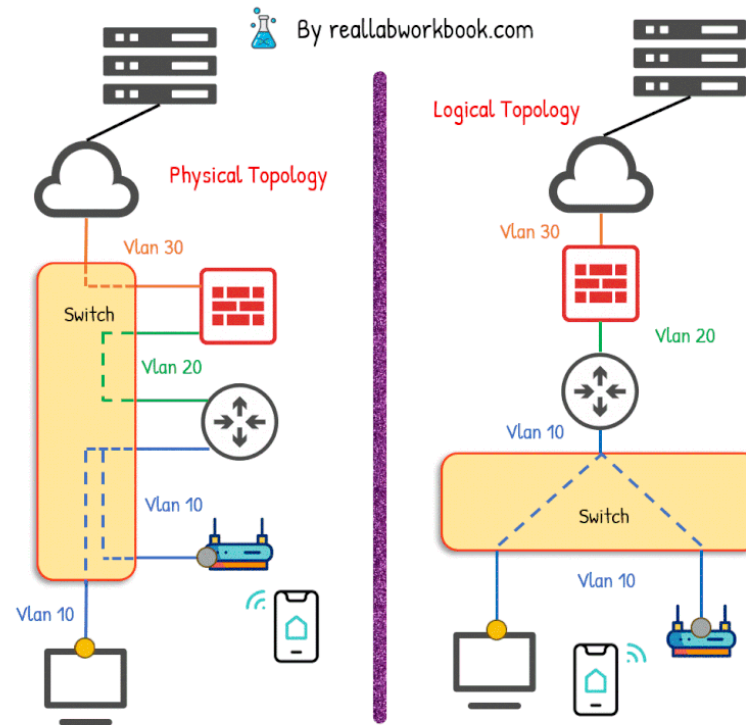
Physical topology is the physical layout of the components on the network.



Logical topology determines how the hosts access the medium to communicate across the network.

CATEGORIES OF NETWORK TOPOLOGY

Physical Topology VS Logical Topology

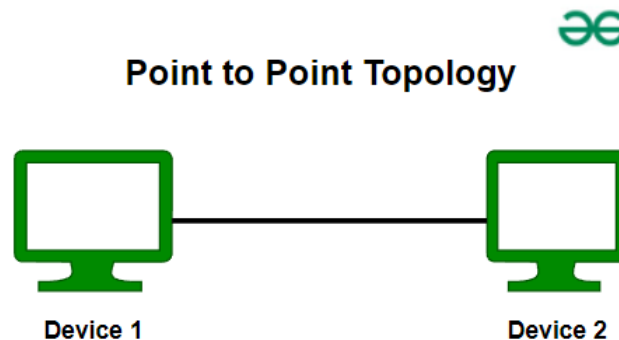


TYPES OF NETWORK TOPOLOGY

- Point-to-Point
- Mesh
- Star
- Bus
- Ring
- Tree
- Hybrid

POINT-TO-POINT TOPOLOGY

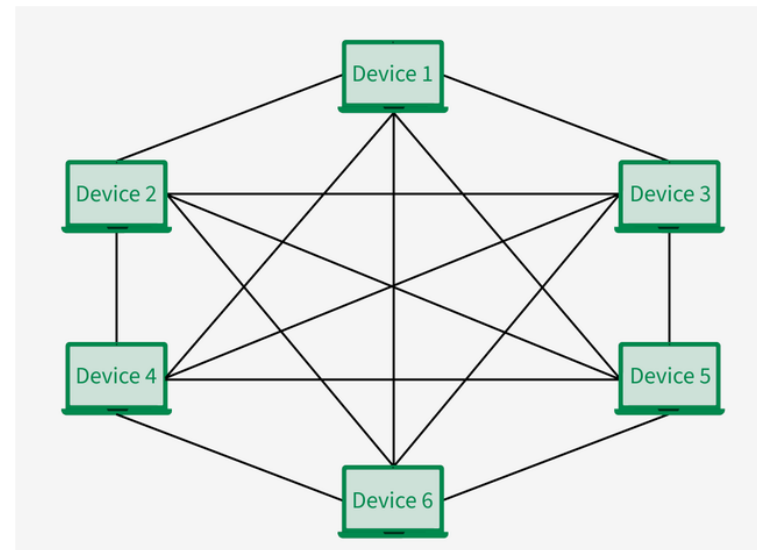
- Direct connection between two nodes
- High bandwidth
- Simple and efficient



Point to Point Topology

MESH TOPOLOGY

- Every node connected to every other node
- Highly reliable and secure
- High cost and complex

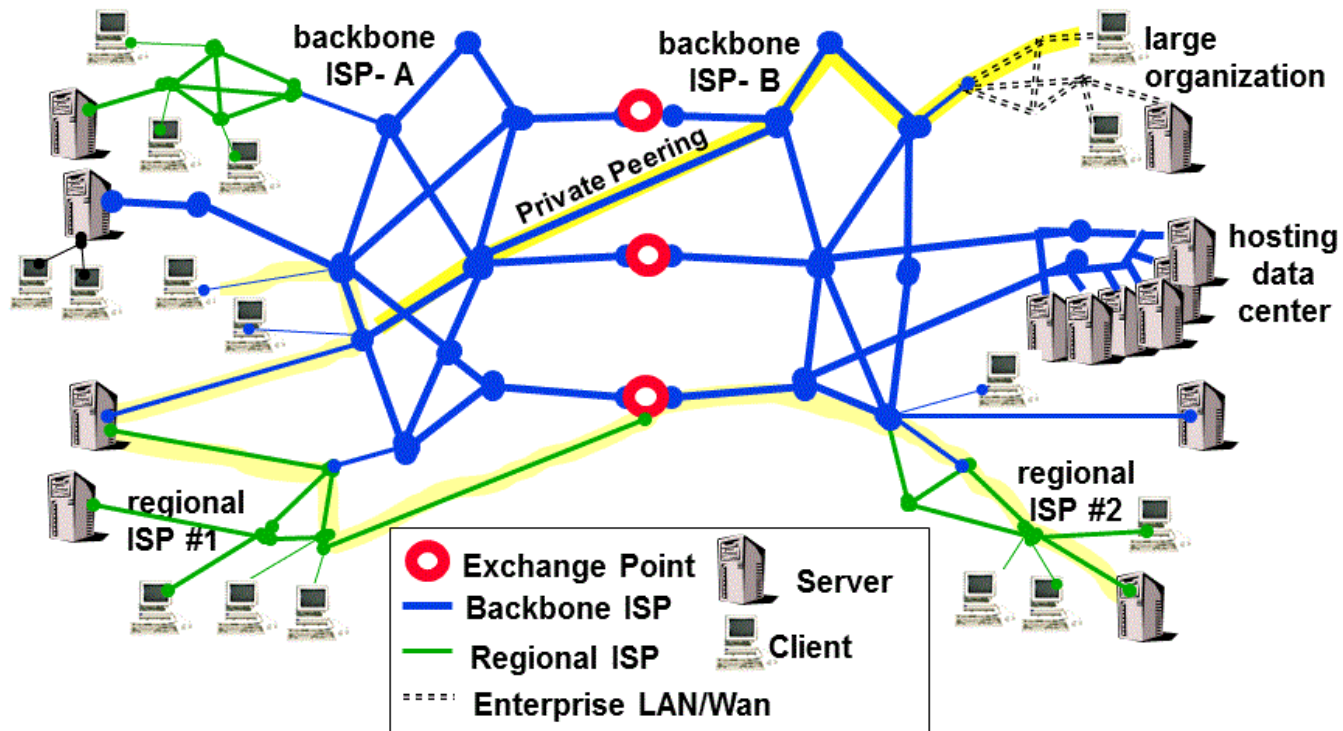


Mesh Topology

Applications in Internet Infrastructure:

- Internet backbone infrastructure connecting ISPs
- Critical infrastructure requiring high availability
- Cloud service provider interconnections
- Military and emergency communication systems

MESH TOPOLOGY



MESH TOPOLOGY

Advantages of Mesh Topology

- Communication is very fast between the nodes.
- Mesh Topology is robust.
- The fault is diagnosed easily. Data is reliable because data is transferred among the devices through dedicated channels or links.
- Provides security and privacy.

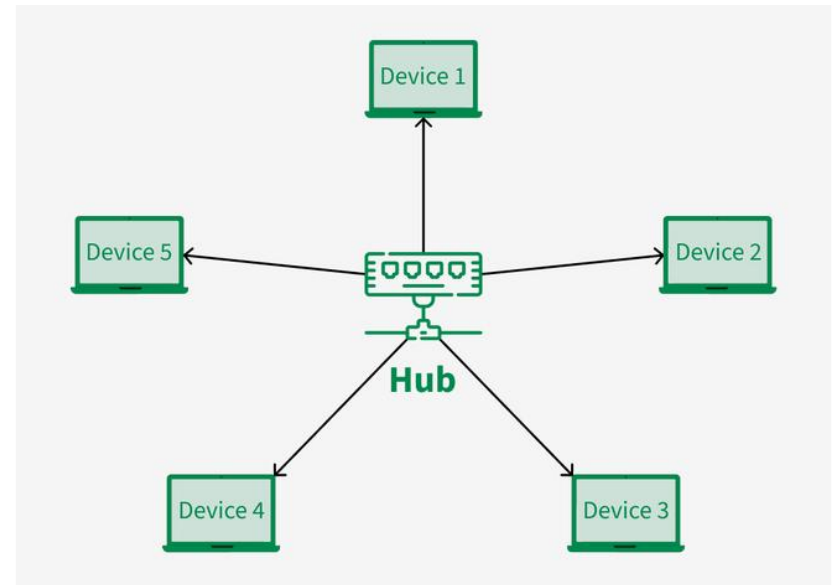
Disadvantages of Mesh Topology

- Installation and configuration are difficult.
- The cost of cables is high as bulk wiring is required, hence suitable for less number of devices.
- The cost of maintenance is high.

A common example of mesh topology is the **internet backbone**, where various internet service providers are connected to each other via dedicated channels. This topology is also used in military communication systems and aircraft navigation systems.

STAR TOPOLOGY

- All nodes connected to a central hub
- Easy to install and manage
- Hub failure affects network



Star Topology

Applications in Internet Infrastructure:

- Local Area Networks (LANs) in offices and homes
- Data centers with centralized switching infrastructure
- Wireless access points connecting multiple devices
- Enterprise networks with managed switches
- Content Delivery Networks (CDNs) at distribution points

STAR TOPOLOGY EXAMPLE

What is
**Content Delivery
Network (CDN)**
in System Design



STAR TOPOLOGY

Advantages of Star Topology

- If N devices are connected to each other in a star topology, then the number of cables required to connect them is N. So, it is easy to set up.
- Each device requires only 1 port i.e. to connect to the hub, therefore the total number of ports required is N.
- It is Robust. If one link fails only that link will affect and not other than that.
- Easy to fault identification and fault isolation.
- Star topology is cost-effective as it uses inexpensive coaxial cable.

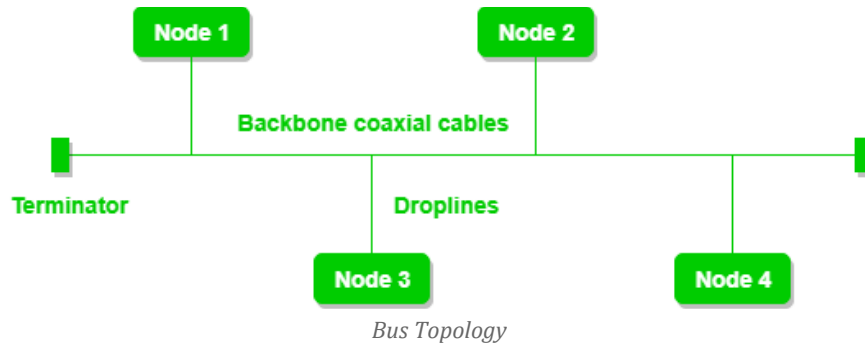
Disadvantages of Star Topology

- If the concentrator (hub) on which the whole topology relies fails, the whole system will crash down.
- The cost of installation is high.
- Performance is based on the single concentrator i.e. hub.

A common example of star topology is a **local area network (LAN)** in an office where all computers are connected to a central hub. This topology is also used in wireless networks where all devices are connected to a wireless access point.

BUS TOPOLOGY

- Single backbone cable
- Low cost
- Backbone failure crashes network



Applications in Internet Infrastructure:

- Early Ethernet networks (10BASE2 and 10BASE5)
- Small office or home networks with limited devices
- Backbone connections in some legacy systems

BUS TOPOLOGY

Advantages of Bus Topology

- If N devices are connected to each other in a bus topology, then the number of cables required to connect them is 1, known as the backbone cable, and N drop lines are required.
- Coaxial or twisted pair cables are mainly used in bus-based networks that support up to 10 Mbps.
- The cost of the cable is less compared to other topologies, but it is used to build small networks.
- CSMA is the most common method for this type of topology.

Disadvantages of Bus Topology

- A bus topology is quite simpler, but still, it requires a lot of cabling.
- If the common cable fails, then the whole system will crash down.
- Adding new devices to the network would slow down the network.
- Security is very low.

A common example of a bus topology is the Ethernet LAN, where all devices are connected to a single coaxial cable or twisted-pair cable. This topology is also used in **cable television networks**.

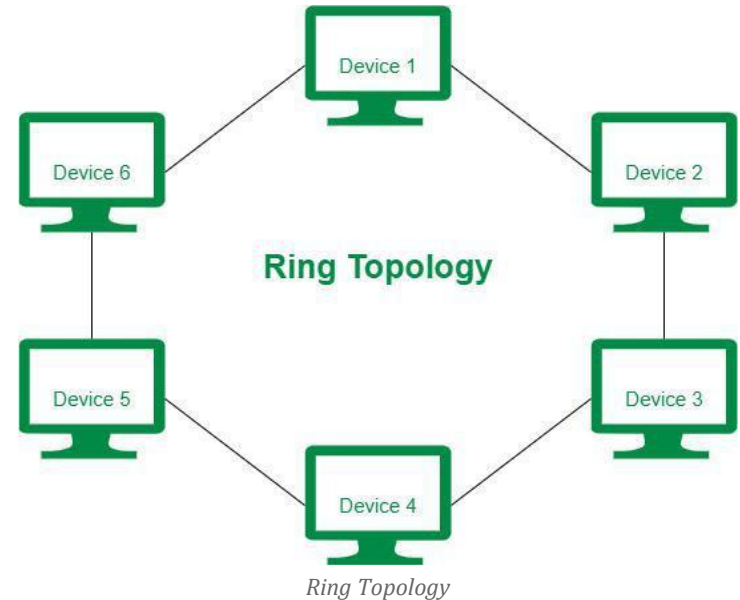
RING TOPOLOGY

- Nodes connected in circular fashion
- Token passing mechanism
- Failure of one node affects network

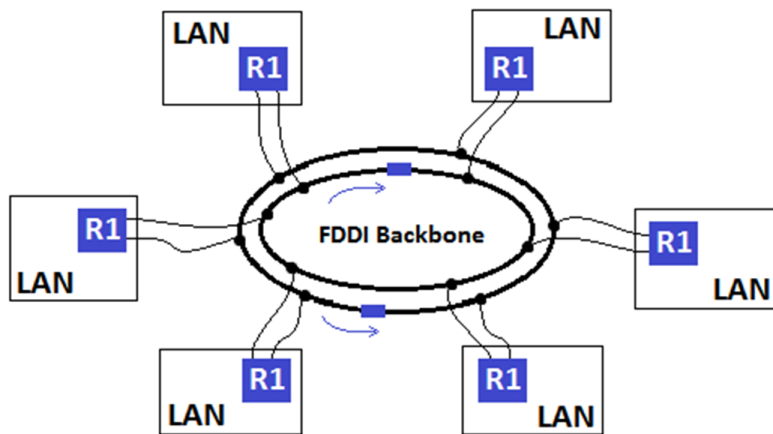
Dual Ring Topology

Applications in Internet Infrastructure:

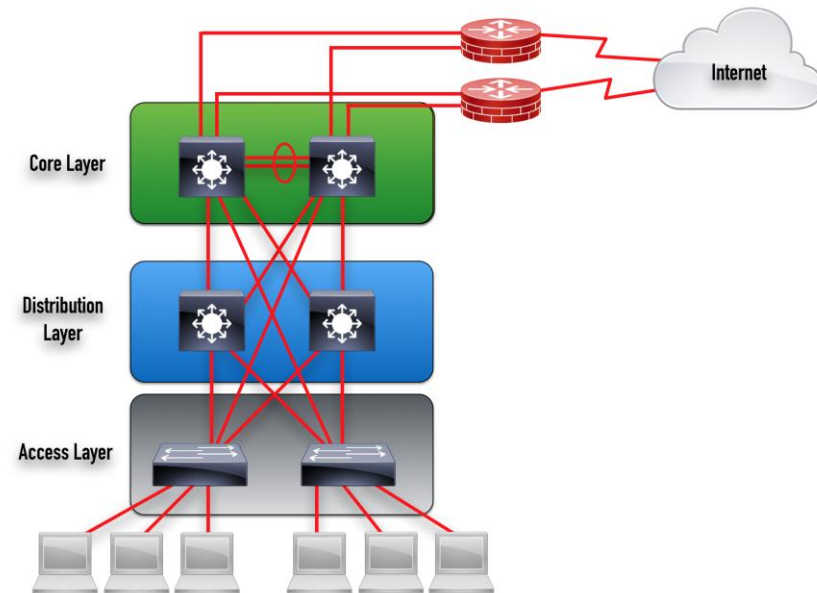
- Fiber Distributed Data Interface (FDDI) networks
- Token Ring networks in legacy enterprise systems
- Redundant backbone connections between data centres.



RING TOPOLOGY EXAMPLE



FDDI networks



Data Center Topology

RING TOPOLOGY

The most common access method of ring topology is token passing.

Token passing: It is a network access method in which a token is passed from one node to another node.

Token: It is a frame that circulates around the network.

Operations of Ring Topology

- One station is known as a **monitor** station which takes all the responsibility for performing the operations.
- To transmit the data, the station has to hold the token. After the transmission is done, the token is to be released for other stations to use.
- When no station is transmitting the data, then the token will circulate in the ring.
- There are two types of token release techniques: **Early token release:** releases the token just after transmitting the data and
- **Delayed token release:** releases the token after the acknowledgment is received from the receiver.

RING TOPOLOGY

Advantages of Ring Topology

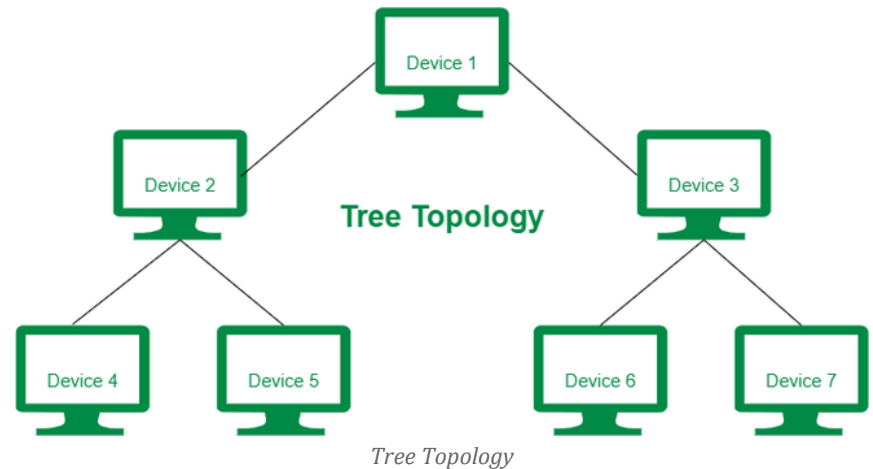
- The data transmission is high-speed.
- The possibility of collision is minimal in this type of topology.
- Cheap to install and expand.
- It is less costly than a star topology.

Disadvantages of Ring Topology

- The failure of a single node in the network can cause the entire network to fail.
- Troubleshooting is difficult in this topology.
- The addition of stations in between or the removal of stations can disturb the whole topology.
- Less secure.

TREE TOPOLOGY

- Hierarchical structure
- Scalable
- Central hub failure affects the network

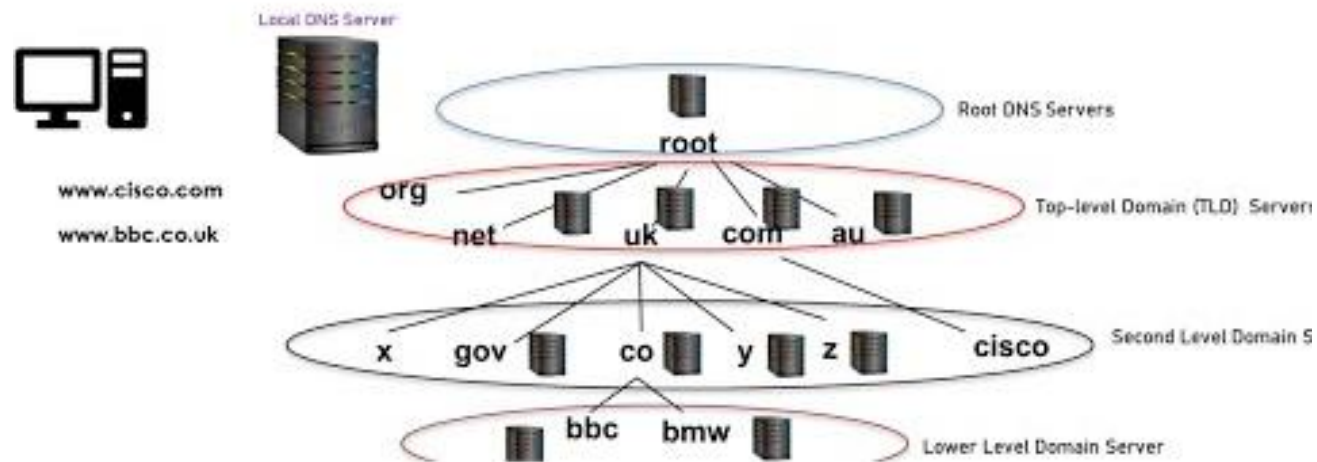


Applications in Internet Infrastructure:

- Large enterprise networks spanning multiple buildings
- University campus networks
- ISP distribution networks from central offices to neighborhoods
- Domain Name System (DNS) hierarchical structure

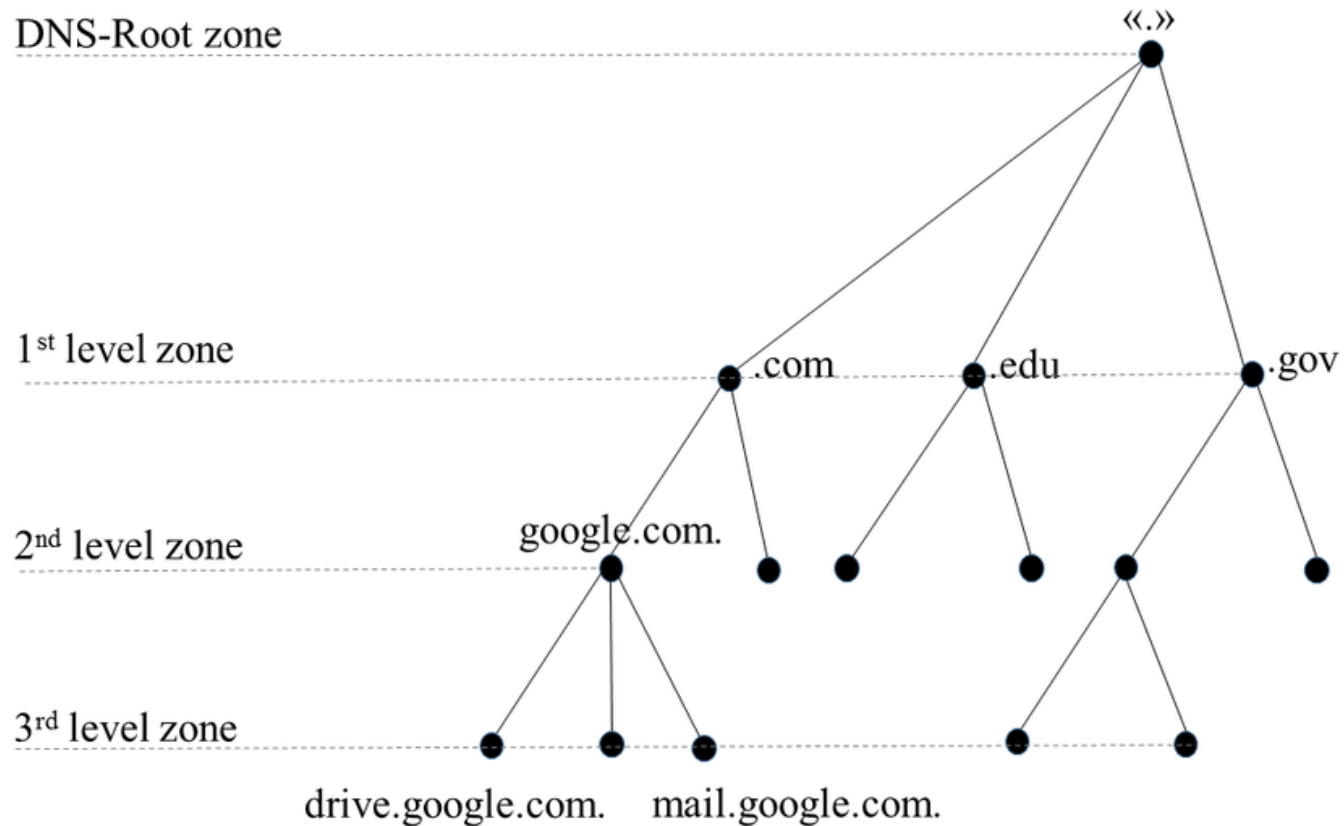
TREE TOPOLOGY

Domain Name System (DNS) hierarchical structure



TREE TOPOLOGY

Domain Name System (DNS) hierarchical structure



TREE TOPOLOGY

Advantages of Tree Topology

It allows more devices to be attached to a single central hub, thus it decreases the distance that the signal travels to come to the devices.

It allows the network to get isolated and also prioritise from different computers.

We can add **new devices to the existing network**.

Error detection and **error correction** are very easy in a tree topology.

Disadvantages of Tree Topology

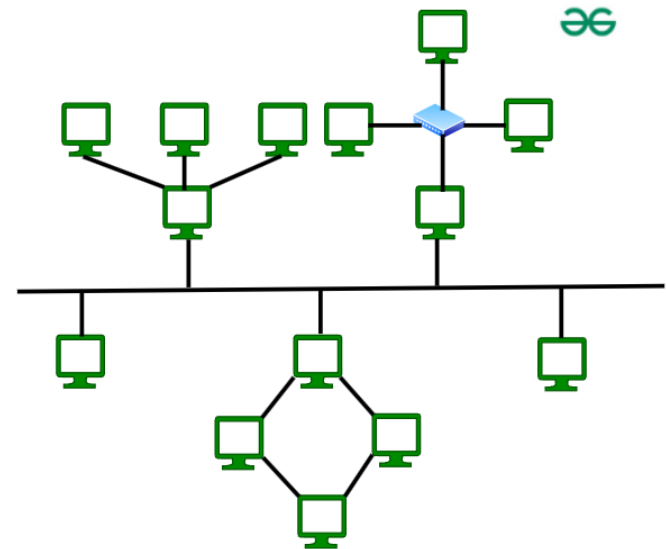
If the central hub gets fails the entire system fails.

The cost is high because of the cabling.

If new devices are added, it becomes difficult to reconfigure.

HYBRID TOPOLOGY

- Combination of multiple topologies
- Flexible and scalable
- Complex and costly



Hybrid Topology

HYBRID TOPOLOGY

Advantages of Hybrid Topology

- This topology is **very flexible** .
- The size of the network can be easily expanded by **adding new devices**.

Disadvantages of Hybrid Topology

- It is challenging **to design the architecture** of the Hybrid Network.
- **Hubs** used in this topology are **very expensive**.
- The infrastructure cost is very high as a hybrid network **requires a lot of cabling and network devices** .

A common example of a hybrid topology is a university campus network.

NETWORK TOPOLOGIES

Considerations for Internet Applications

When selecting or analyzing network topologies for internet applications, several factors must be considered:

- **Scalability:** The ability to add new devices and expand the network without significant redesign
- **Redundancy:** Multiple paths for data to ensure continued operation during failures
- **Performance:** Bandwidth capacity, latency, and throughput requirements
- **Cost:** Initial investment in hardware, cabling, and ongoing maintenance expenses
- **Management complexity:** Ease of monitoring, troubleshooting, and administration
- **Security:** Ability to implement access controls and monitor traffic flow

The modern internet relies primarily on mesh topology at its core for inter-ISP connections and routing, with star and tree topologies at the edge for connecting end-users.

IMPORTANCE OF NETWORK TOPOLOGY

- Improves performance
- Enhances reliability
- Enables scalability
- Strengthens security



CONCLUSION

Choosing the right network topology is essential for efficient, secure, and scalable network design.